PREP-RASHAN

A Unit of Prep-Rashan Tutorial

DAILY CURRENT AFFAIRS - RASHAN

Comprehensive Compilation for UPSC Civil Services Examination

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General Studies Paper I • Paper II • Paper III • Paper IV

Keywords Defined • PYQ (Prelims & Mains) • Practice Questions (Prelims & Mains)

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GENERAL STUDIES PAPER I

[GS-I] Indian Heritage & Culture, History, Geography of the World and Society

GS-I | Art & Culture – Literature and Documentary Heritage

1. Rajatarangini, Mahabharata and Arthashastra Enter UNESCO's Asia-Pacific Memory Register

Syllabus: *GS-I: Indian culture – salient aspects of literature from ancient to modern times. GS-II: International institutions (UNESCO).*

CONTEXT

Three classics of Indian thought – **Kalhana's Rajatarangini**, the **Mahabharata** and **Kautilya's Arthashastra** – entered UNESCO's **Memory of the World** list for the **Asia-Pacific region**, decided at the **11th MOWCAP** meeting held in **Jakarta**.



Illustration: an old manuscript and a bound book on a stone pedestal under a glowing lamp – heritage kept safe for the future.

KEYWORDS DEFINED

Keyword	Meaning
Memory of the World (MoW)	A **UNESCO programme (1992)** to **protect and open up documentary heritage** – manuscripts, archives, recordings and rare texts.
MOWCAP	The **Asia-Pacific regional committee** of the Memory of the World programme; it keeps the **regional register**.
Taranga	A **'wave'** – each of the **eight books** of the Rajatarangini.
Saptanga	Kautilya's **seven limbs of the state** – ruler, ministers, territory, fort, treasury, army and ally.
Shadgunya	The **six-fold foreign policy** of the Arthashastra – peace, war, neutrality, march, alliance and double policy.

The Memory of the World programme

- Set up by **UNESCO in 1992** to save **documentary heritage** threatened by **neglect, decay, war and loss**.

- **Three registers: International, Regional and National**, by the reach of a document's significance.
- Nominations are judged by **expert panels** on **authenticity, importance, rarity, influence** and **state of preservation**.
- Promotes **conservation, microfilming and digitisation** so that fragile texts stay **accessible**.

Rajatarangini – 'River of Kings'

- A **Sanskrit chronicle** by **Kalhana**, written about **1148–1150 CE** under King **Jayasimha** of **Kashmir**.
- Traces **Kashmir's kings and dynasties** across **eight Tarangas**.
- Kalhana used **older chronicles, land grants, inscriptions and coins** – an early example of **historical method** in India.
- Records **revolts, famines, taxes, temple patronage** and religious life of medieval Kashmir.

Mahabharata and Arthashastra

- **Mahabharata:** credited to **Vyasa**; grew from **Jaya (about 8,800 verses)** to **Bharata (about 24,000)** and then the full epic.
- Tells of the **Kurukshetra war** of **Pandavas and Kauravas** and the four aims – **Dharma, Artha, Kama, Moksha**.
- The **700-verse Bhagavad Gita** sits in the **Bhishma Parva**; the epic shaped **Yakshagana, Kathakali** and **Wayang**.
- **Arthashastra:** linked to **Kautilya (Chanakya/Vishnugupta)** of **Chandragupta Maurya's** court; **15 books** on administration, tax, law and treasury.
- **Rediscovered by R. Shamasastry in 1905**, published **1909**; gives the **Saptanga theory**, **Mandala theory**, **Shadgunya** and **Sama, Dana, Bheda, Danda**.

Why it matters

- Puts India's **historical, epic and political** writing on a **global heritage map**.
- Supports **digitisation and access** for scholars and the public.
- **Value addition:** India's entries on the **International MoW Register** include the **Rigveda manuscripts (2007)** and the **Gilgit Manuscripts (2017)**; the **Ramcharitmanas, Panchatantra and Sahrdayaloka-Locana** entered the **Asia-Pacific register in 2024** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **ancient texts and authors**, **UNESCO lists** and Arthashastra concepts like **Saptanga**.

PYQ – MAINS

- **Pattern:** Mains asks how **literature reflects society and statecraft** and about **preserving heritage**.

FQ – PRELIMS (Possible Question)

- Which of the following was written by Kalhana? (a) Arthashastra (b) Rajatarangini (c) Harshacharita (d) Mudrarakshasa **(Ans: b)**

FQ – MAINS (Possible Question)

- "The Rajatarangini marks the beginning of historical writing in India." Examine. (150 words)

GS-I | Modern Indian History – Revolt of 1857

2. Ambala's Memorial to the Martyrs of 1857

Syllabus: *GS-I: Modern Indian history – significant events and personalities; the freedom struggle.*

CONTEXT

A large **memorial to the martyrs of the 1857 First War of Independence** was opened in **Ambala Cantonment, Haryana**, built for about **₹594 crore** on **22 acres**.



Illustration: a memorial with a tall pillar, galleries and an open-air theatre beside a highway, with an old rifle and cartridge in an inset.

KEYWORDS DEFINED

Keyword	Meaning
Enfield cartridge	The **greased cartridge** of the new **Enfield rifle**, rumoured to use **cow and pig fat** – the spark of 1857.
Doctrine of Lapse	**Dalhousie's** rule: a state whose ruler died **without a natural heir** passed to the **Company**.
Subsidiary Alliance	**Wellesley's** system binding Indian rulers to keep **British troops** at their own cost.
Government of India Act, 1858	Ended **Company rule** and placed India **under the British Crown**.
Phulkari	A traditional **embroidery of Punjab and Haryana**, used in the memorial's design.

The memorial

- Located on the **Grand Trunk Road** in **Ambala Cantonment**, an old **military station** that saw **sepoy unrest in 1857**.
- **22 exhibition galleries**, an **open-air theatre for 2,500** and a **light-and-sound show**.
- Mixes local crafts such as **Phulkari** and **Sarkanda** with **virtual reality, holograms and projection mapping**.
- Tells the story of **lesser-known fighters** and the **timeline** of the revolt.

Why the revolt broke out

- **Immediate cause:** the **greased cartridges** of the **Enfield rifle**.
- **Political: Doctrine of Lapse** and the **annexation of Awadh (1856)** angered rulers, landlords and soldiers.
- **Economic:** heavy land revenue, **ruined handicrafts** and rural distress.
- **Military:** low pay, poor promotion and **service overseas** rules for sepoys.

Course and leaders

- **Mangal Pandey** at **Barrackpore (March 1857)**; the main rising began at **Meerut on 10 May 1857**.
- **Delhi: Bahadur Shah Zafar** as symbolic head, **Bakht Khan** in military command.
- **Kanpur: Nana Saheb**, **Tatya Tope**, **Azimullah Khan**; **Jhansi: Rani Lakshmibai**; **Lucknow: Begum Hazrat Mahal**.
- **Bihar: Kunwar Singh** of **Jagdishpur**; **Faizabad: Maulvi Ahmadullah Shah**; **Bareilly: Khan Bahadur Khan**.
- **Haryana: Rao Tula Ram** of **Rewari** and **Raja Nahar Singh** of **Ballabhgarh**; **Allahabad: Maulvi Liaquat Ali**.

Results

- The **Government of India Act, 1858** ended **East India Company rule**; power passed to the **Crown**.
- The army was **reorganised**; annexation policies were dropped.
- **Value addition: V.D. Savarkar** called it **'The Indian War of Independence' (1909)**; **R.C. Majumdar** saw it as neither first, national nor a war of independence *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims pairs **1857 leaders with centres** and asks about **causes** and the **1858 Act**.

PYQ – MAINS

- **Pattern:** Mains asks whether 1857 was a **mutiny or a war of independence** and about its **consequences**.

FQ – PRELIMS (Possible Question)

- Match: 1. Kunwar Singh – Jagdishpur 2. Begum Hazrat Mahal – Lucknow 3. Khan Bahadur Khan – Kanpur. How many pairs are correct? **(Ans: Only two)**

FQ – MAINS (Possible Question)

- "The Revolt of 1857 failed, yet it changed the nature of British rule." Discuss. (150 words)

GS-I | Geography – Places in News (Mapping); Energy Security

3. Yanbu: Saudi Arabia's Red Sea Oil Gateway Under Threat

Syllabus: *GS-I: Important geographical features and resources (mapping). GS-II: Effect of policies of other countries on India's interests. GS-III: Energy security.*

CONTEXT

Saudi Arabia warned of **serious threats** to its **Yanbu oil export hub on the Red Sea** after repeated **missile and drone attacks** by Yemen's **Houthi** group.



Illustration: an oil export terminal on a desert coast with tankers at deep-water jetties, refinery towers, storage tanks and a pipeline from inland.

KEYWORDS DEFINED

Keyword	Meaning
Petroline	Saudi Arabia's **East–West crude pipeline** (about **1,200 km**) from the **Gulf side to Yanbu**.
Strait of Hormuz	The narrow **outlet of the Persian Gulf** – the world's busiest **oil chokepoint**.
Bab-el-Mandeb	The strait linking the **Red Sea to the Gulf of Aden**, between **Yemen and Djibouti/Eritrea**.
Houthis	A **Yemeni armed movement** (Ansar Allah) controlling much of **north-west Yemen**.
bpd	**Barrels per day** – the usual measure of **oil flow or refining capacity**.

Where Yanbu is (mapping)

- A port and **industrial city** in **Al Madinah Province**, on the **eastern shore of the Red Sea**.
- About **300 km north of Jeddah** and **160 km south-west of Medina**.
- Built up from the **mid-1970s** by the **Royal Commission for Jubail and Yanbu**; **Jubail** is its twin on the **Gulf coast**.

What it does

- **End point of the Petroline:** carries up to **5 million barrels a day** of crude from the **east** to the **Red Sea**.
- **Refining:** the **YASREF refinery (about 400,000 bpd)**, **Aramco** units and **SABIC** petrochemical plants.

- **King Fahd Industrial Port** with **deep-water berths** for big tankers.
- New **Vision 2030** projects: **green hydrogen, minerals, desalination**, links to **NEOM**.

Why it was built

- Planned during the **Iran–Iraq War** and **'Tanker War' (1980s)** as a route that **avoids the Gulf and Hormuz**.
- Lets Saudi oil reach **Europe and the Mediterranean** via the **Red Sea and Suez Canal**.
- **Map tip: Red Sea** neighbours – **Egypt, Sudan, Eritrea, Djibouti** (west); **Saudi Arabia, Yemen** (east); **Jordan, Israel** at the **Gulf of Aqaba** (north).

Why it matters for India

- Attacks could **raise crude prices, freight and insurance**, hitting India's **import bill and inflation**.
- Saudi Arabia is one of India's **top crude suppliers**; a **Hormuz bypass** that fails weakens supply security.
- **Value addition:** India's **Strategic Petroleum Reserves** at **Visakhapatnam, Mangaluru and Padur** hold about **5.33 million tonnes** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims mapping asks about **Red Sea neighbours, straits** (Bab-el-Mandeb, Hormuz) and **ports** in news.

PYQ – MAINS

- **Pattern:** Mains asks about **West Asian conflicts** and **India's energy security**.

FQ – PRELIMS (Possible Question)

- Yanbu, recently in the news, lies on the coast of: (a) the Persian Gulf (b) the Red Sea (c) the Arabian Sea (d) the Gulf of Oman **(Ans: b)**

FQ – MAINS (Possible Question)

- "Chokepoint risks in West Asia make energy diversification a strategic need for India." Discuss. (150 words)

GENERAL STUDIES PAPER II

[GS-II] Governance, Constitution, Polity, Social Justice & International Relations

GS-II | Polity – Fundamental Rights; Child Rights & Digital Safety

4. Children's Right to the Internet: Safety Without Blanket Bans

Syllabus: *GS-II: Fundamental rights; judiciary; welfare of vulnerable sections (children); issues in governance of digital platforms. GS-III: Cyber security.*

CONTEXT

The **Supreme Court**, ruling in **Just Rights for Children Alliance v. S. Harish (2026)**, held that **children have a fundamental right to access and use the internet**, rejected **blanket age-based bans**, and asked the Centre to build **shared hash and URL databases** against **child sexual abuse material (CSAM)**.



Illustration: a child using a tablet inside a protective shield, with a padlock, heart and safety icons, while dark warning symbols bounce off.

KEYWORDS DEFINED

Keyword	Meaning
CSAM	**Child sexual abuse material** – any visual depiction of the **sexual abuse of a child**; the Court prefers this term to 'child pornography'.
Perceptual hash	A **digital fingerprint** of an image or video that still **matches after small edits** – used to spot **known abuse material**.
Safety by design	Building **protections into apps and platforms** from the start rather than relying only on **bans**.
Article 19(1)(a)	**Freedom of speech and expression**, which includes the **right to receive and share information**.
Digital access as life	Courts have read **meaningful digital access** into **Article 21** (right to life).

What the Court said

- **Internet access** for children is part of their **fundamental rights**; safety and freedom are **not opposites**.

- **Blanket age bans** were rejected – they may push children to **unregulated, riskier sites** and make them **afraid to report abuse**.
- **Age-appropriate limits and safeguards** are preferred to **denial of access**.

Constitutional roots

- **Article 19(1)(a):** the **right to know** – **Cricket Association of Bengal (1995)** tied speech to **receiving and sharing information**.
- **Anuradha Bhasin (2020):** online **speech** and **trade** protected under **Articles 19(1)(a) and 19(1)(g)**.
- **Pragya Prasun (2025):** shutting **persons with disabilities** out of **digital KYC** violated **Article 21**.
- **Balance:** the Constitution limits **both State power and individual liberty** – too much control risks **authoritarianism**, too little risks **disorder**.

The risks and the gaps

- **Cyberbullying, privacy breaches, fraud** and **obscene content**.
- **CSAM** spreading through **paid promotions and redirect links**.
- **Complaint-based takedown:** removal depends largely on **individual complaints**.
- **No cross-platform matching:** material removed on one site is **not flagged on others**.
- **Slow police response:** reports routed via a **US-based non-profit** first; **Google and Meta** have now agreed to report **directly to Indian agencies**.

Way forward

- **National hash and URL databases** to catch the **same material across platforms**.
- **Verify before acting:** no takedown or database entry on **a single complaint**; a **grievance route** for wrong matches; databases used **only for CSAM**.
- **Online-safety lessons from primary school**; parents, schools, platforms and government **share responsibility**.
- **World view: Australia** bars social media accounts **under 16**; **Brazil** links children's accounts to a **guardian**; **Andhra Pradesh and Karnataka** have discussed **screen-time limits**.
- **Value addition:** the **POCSO Act, 2012** and **Section 67B of the IT Act, 2000** penalise CSAM; the **DPDP Act, 2023** needs **verifiable parental consent** for processing children's data *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **fundamental rights cases**, the **IT Act**, **POCSO** and **data protection** rules.

PYQ – MAINS

- **Pattern:** Mains asks how to **balance rights with regulation** in cyberspace and protect **vulnerable groups**.

FQ – PRELIMS (Possible Question)

- Consider: 1. The Supreme Court upheld a blanket ban on internet use by children under 16. 2. Perceptual hashing can match images even after small edits. 3. The DPDP Act, 2023 requires verifiable parental consent for children's data. Which are correct? **(Ans: 2 and 3 only)**

FQ – MAINS (Possible Question)

- "Protecting children online should not mean shutting them out." Discuss with reference to recent judicial thinking. (250 words)

GS-II | Social Sector – Health; Drug Pricing

5. Cheaper Cancer Drugs on Paper: Do Patients Really Save?

Syllabus: *GS-II: Issues relating to development and management of health services; government policies and interventions.*

CONTEXT

The **NPPA** capped **trade margins at 30%** on **137 non-scheduled anti-cancer drugs**, cutting the **MRP of about 900 brands** – but lower printed prices **do not always mean real savings** for patients.



Illustration: medicines on a weighing scale against hospital bills and coins, with a falling price tag and a cancer-care ribbon.

KEYWORDS DEFINED

Keyword	Meaning
NPPA	**National Pharmaceutical Pricing Authority (1997)** – fixes and monitors **drug prices** under the **Department of Pharmaceuticals**.
DPCO, 2013	**Drugs (Prices Control) Order** – caps prices of **scheduled (essential) medicines**; others may raise prices by **up to 10% a year**.
Trade margin	The gap between the price at which the **maker sells (to the first buyer)** and the **MRP** paid by the patient.
Non-scheduled drug	A medicine **outside the price-control schedule** (NLEM-based) of the DPCO.
Out-of-pocket expenditure	Health spending paid **directly by the patient** at the time of care.

The burden of cancer care

- Typical treatment costs about **₹4.1 lakh**, of which about **₹3.3 lakh (80%)** is paid **out of pocket**.
- About **40% of patients borrow or pledge assets** to pay for care.
- **Five-year breast-cancer survival:** about **65.5% in India** vs about **90%** in rich countries.
- Advanced care is **concentrated in private hospitals**; **tests, surgery, stays and travel** add to drug costs.

What the price cut did

- Of **1,024 brands** studied, **45.4%** saw cuts **above 50%**, **19.6% above 70%** and **0.8% above 90%**.
- Only **16.5%** saved patients **more than ₹10,000** – big percentages, **small rupee gains**.

- **Palbociclib:** one brand fell from about **₹40,000 to ₹2,720**, but **767 other brands** already sold at **₹2,800–9,000**.
- **About 194 brands** (priced **₹1–10 lakh**) saw **no cut**; **ribociclib** and **pembrolizumab** were unchanged.
- **Earlier step:** margins on **42 anti-cancer drugs** were rationalised in **2019**.

Why savings leak

- A **flat 30% margin** earns a hospital more on an **₹8,000** drug than on a **₹2,000** one – a nudge to **stock costly brands** over **generics**.
- Patients have **little bargaining power** and **little price information**.
- **No independent price database** to check prices across **makers, stockists, pharmacies and hospitals**.
- **Hospital charges are unregulated**, unlike **telecom or electricity**; agencies **buy separately**, losing **bulk discounts**.

Way forward

- **Monitor real prices** before and after caps with **State drug regulators**; build a **public pricing database**.
- **Itemised hospital bills**; action on **overcharging**; **pooled procurement** by agencies and insurers.
- **Cost-based caps** linked to **standard treatment protocols**; a **National Cancer Mission** for public capacity and second opinions.
- **Full-journey cover** under **Ayushman Bharat** and State schemes; **quality generics** first; protect **R&D incentives**; pay by **patient outcomes**.
- **Value addition: Jan Aushadhi Kendras** sell generics; **AMRIT pharmacies** offer cancer drugs at large discounts; the **NLEM 2022** has **384 medicines** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **NPPA, DPCO, NLEM** and health schemes like **PM-JAY** and **Jan Aushadhi**.

PYQ – MAINS

- **Pattern:** Mains asks about **out-of-pocket spending**, **affordable medicines** and **universal health coverage**.

FQ – PRELIMS (Possible Question)

- Consider: 1. NPPA works under the Ministry of Health and Family Welfare. 2. Non-scheduled formulations may raise prices by up to 10% a year under DPCO 2013. 3. Trade margins on 42 anti-cancer drugs were rationalised in 2019. Which are correct? **(Ans: 2 and 3 only)**

FQ – MAINS (Possible Question)

- "Lower MRPs do not by themselves make cancer care affordable." Examine and suggest reforms. (250 words)

GS-II | Polity – Constitutional Bodies (State Public Service Commission)

6. Removing a State PSC Chairman: Who Can and How

Syllabus: *GS-II: Appointment to various constitutional posts, powers, functions and responsibilities of constitutional bodies.*

CONTEXT

A delegation met the **President** seeking action on a recommendation to **remove the Chairman and members of the Kerala Public Service Commission** – a power the Constitution gives **only to the President**.



Illustration: a commission building with pillars, an exam hall, a balance scale, a gavel and an order file.

KEYWORDS DEFINED

Keyword	Meaning
SPSC	**State Public Service Commission** – the **constitutional recruiting body** for a State's services.
Part XIV	The Constitution's part on **services under the Union and States**; **Articles 315–323** cover the PSCs.
Article 317	Lays down **removal and suspension** of PSC members – by the **President**, with a **Supreme Court inquiry** for misbehaviour.
Charged expenditure	Spending on the **Consolidated Fund** that the legislature **cannot vote down** – covers PSC salaries.
JSPSC	A **Joint State PSC** for two or more States, created by **Parliament** on their request.

Constitutional basis and make-up

- **Articles 315–323, Part XIV**; a **constitutional body**.
- **Chairman and members** are **appointed by the Governor**; the Constitution does **not fix their number** or **qualifications**.
- **At least half** the members must have served **10 years** under the Union or a State government.
- **Term: six years or till age 62**, whichever is earlier; the Governor fixes **service conditions** and may name an **acting Chairman**.

Removal – the key fact

- Appointed by the **Governor**, but **removable only by the President** (**Article 317**).
- **Grounds: insolvency**, **paid outside employment**, **infirmity of mind or body**, or **misbehaviour**.
- For **misbehaviour**, the President must refer the case to the **Supreme Court**; its advice is **binding**.
- The **Governor can suspend** a member while the **Supreme Court inquiry** is on.
- A member may **resign** by writing to the **Governor**.

Independence safeguards

- **Security of tenure** and removal only by the **President**.
- Service conditions **cannot be changed to their disadvantage** after appointment.
- **Expenses charged** on the **Consolidated Fund of the State**.
- **After office:** the Chairman may become **Chairman or member of UPSC or another SPSC**, but **no other government job**; a member may also move **only within UPSC/SPSCs**.

Functions

- Conducts **exams for State services**; advises on **recruitment, promotion, transfer and discipline**.
- Sends an **annual report to the Governor**, laid before the **State Legislature** with **reasons for not accepting** its advice.
- **Value addition:** a **JSPSC** is created by **Parliament**; the **UPSC can serve a State** if the **Governor requests** and the **President agrees** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks **who appoints and who removes** constitutional office holders and their **terms**.

PYQ – MAINS

- **Pattern:** Mains asks about the **independence and credibility** of recruiting bodies amid **paper leaks and delays**.

FQ – PRELIMS (Possible Question)

- The Chairman of a State Public Service Commission can be removed by: (a) the Governor (b) the Chief Minister (c) the President (d) the State Legislature by a resolution **(Ans: c)**

FQ – MAINS (Possible Question)

- "The credibility of State Public Service Commissions is central to good governance." Discuss the safeguards and reforms needed. (150 words)

GS-II | Governance – Internal Security; India–Nepal–Bhutan Relations

7. New Border Travel Rules for India, Nepal and Bhutan

Syllabus: *GS-II: India and its neighbourhood; government policies. GS-III: Security challenges and their management in border areas.*

CONTEXT

The **Ministry of Home Affairs** notified the **Immigration and Foreigners (Exemption) (Amendment) Order, 2026**, changing the **2025 Order** to tighten **identity checks** on the **India–Nepal and India–Bhutan** routes while keeping **visa-free travel** for eligible citizens.



Illustration: a mountain border post with a barrier gate, a traveller showing an ID card to an officer, a passport and a plane above.

KEYWORDS DEFINED

Keyword	Meaning
Visa Manual	The Centre's **official compendium of visa and immigration rules**, now **defined in the Order**.
Immigration check post (ICP)	A **designated entry–exit point** where travellers are **checked** by immigration officers.
Third-country national	A person who is **neither Indian nor from the neighbouring country** concerned.
Open border	A border crossed **without passport or visa** – India–Nepal under the **1950 Treaty of Peace and Friendship**.
Immigration and Foreigners Act, 2025	New law that **replaced four older laws** on passports, foreigners and registration *(general knowledge)*.

What the order is

- An **executive order** of the **MHA**, amending the **Immigration and Foreigners (Exemption) Order, 2025**.
- Sets the **documents, exemptions and routes** for entry, transit and exit.
- **Aim:** better **border security**, stop **third-country infiltration**, but keep **visa-free movement** for neighbours.

Rules for Indians

- Going to or returning from **Nepal or Bhutan** by **land or air**: show **one valid ID** listed in the **Visa Manual** and answer questions.
- Using these borders to travel to **any other country**: a **valid passport** is a must, and only **designated check posts** may be used.

Rules for Nepalese and Bhutanese citizens

- Show **one ID from the Visa Manual** when entering or leaving India by **land or air**.
- The exemption **does not apply** when travelling **to or from a third country**.
- **Passport and Indian visa** needed if they **flew from or transited via China, Macau, Hong Kong or Pakistan**.

Who else is exempt

- **Indian armed forces** on duty (and families on government transport).
- **Sri Lankan Tamil refugees** registered and in India **by 9 January 2015**.
- **Crew of foreign military ships** visiting for repair with a **Military Ship Repair Permit**.
- **Tibetans** who came after **1959** with the right **entry permits** (rules differ before and after **30 May 2003**).
- **Persecuted minorities – Hindus, Sikhs, Buddhists, Jains, Parsis, Christians** – from **Afghanistan, Bangladesh, Pakistan** who entered **by 31 December 2024**.
- **Value addition:** India shares about **1,751 km** of border with **Nepal** and **699 km** with **Bhutan**; **SSB** guards both *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **border-guarding forces**, **neighbouring treaties** and new **laws and orders**.

PYQ – MAINS

- **Pattern:** Mains asks about **open-border management** and **cross-border crime**.

FQ – PRELIMS (Possible Question)

- Under the 2026 amendment, a Nepalese citizen needs a passport and an Indian visa if he: (a) enters India by road from Nepal (b) flies from Kathmandu to Delhi (c) has flown in via Hong Kong (d) travels to Bhutan through India by road **(Ans: c)**

FQ – MAINS (Possible Question)

- "An open border is a bridge of friendship and a channel of risk." Discuss with reference to the India–Nepal border. (150 words)

GENERAL STUDIES PAPER III

[GS-III] Technology, Economic Development, Bio-diversity, Environment, Security & Disaster Management

GS-III | Economy – Industry and Manufacturing

8. Annual Survey of Industries 2024–25: Growth at the Top, Strain at the Base

Syllabus: *GS-III: Indian economy – growth, development and employment; industrial policy; effects of liberalisation on industry.*

CONTEXT

MoSPI released the **Annual Survey of Industries (ASI) 2024–25**: the **corporate manufacturing sector's GVA rose 9.28%** and formal factory jobs crossed **2.10 crore**, but **capital and value are heavily concentrated** in a few large units and States.



Illustration: a big automated factory beside many small workshops, with rising bar charts and simple worker figures.

KEYWORDS DEFINED

Keyword	Meaning
ASI	**Annual Survey of Industries** – MoSPI's main survey of **registered factories**: output, jobs, capital and value added.
GVA	**Gross Value Added** – value of output **minus inputs used up**.
NVA	**Net Value Added** – GVA **minus depreciation** of machines and buildings.
Missing middle	Too **few medium-sized firms** between many tiny units and a handful of giants.
Collection of Statistics Act, 2008	The law under which the ASI is run (**amended 2017**; rules of **2011**).

What the ASI covers

- Factories covered by the **Factories Act, 1948** (Sections 2m(i)/(ii)) – **10+ workers with power** or **20+ without**.

- Also **bidi and cigar units**, **electricity units not registered with the CEA**, and units with **100+ employees** in the States' **Business Register of Establishments**.
- Run under the **Collection of Statistics Act, 2008**.

Headline numbers

- **Corporate sector** (over **89% of manufacturing GVA**): **GVA up 9.28%**.
- **NVA per factory up 6.59%**, but **GVA per person only up 2.25%**.
- **Plant and machinery up 7.25%**; **factories up 3.45%**; **persons engaged up 7.57%** to **over 2.10 crore**.
- **Top job sectors:** food products, textiles, basic metals, motor vehicles, apparel – **over 41% of jobs**.
- **Top GVA States: Maharashtra, Gujarat, Tamil Nadu, Karnataka, Uttar Pradesh – over 54%**.

The concentration problem

- **Small units (P&M up to ₹5 crore): 78.69% of factories**, **34.32% of workers**, but only **10.55% of fixed capital** and **17.61% of NVA**.
- **Big units (P&M ₹10 crore+): 15.84% of factories** hold **87.42% of capital** and give **78.07% of NVA**.
- Units with output **below ₹1 crore** (**18.34%** of factories) make just **0.08% of output** with **negative value added**.
- **Labour productivity lags capital** – wages and skills are not keeping pace.

Way forward

- **Credit-linked capital subsidy** so small factories can buy **CNC machines and digital systems**.
- **Industrial parks in eastern States** (Bihar, Bengal, Odisha, Jharkhand) via **PM Gati Shakti** and **freight corridors**.
- **Apprenticeships (NAPS)** and shop-floor reskilling; **plug-and-play clusters** with shared labs and effluent plants.
- **Value addition:** manufacturing is about **17% of GVA**; the **National Manufacturing Policy (2011)** aimed at **25% of GDP** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks **which body releases which survey** (ASI, PLFS, IIP) and what **GVA/NVA** mean.

PYQ – MAINS

- **Pattern:** Mains asks why **manufacturing jobs lag** and about **MSME growth constraints** and **regional imbalance**.

FQ – PRELIMS (Possible Question)

- Consider: 1. The ASI is conducted by MoSPI under the Collection of Statistics Act, 2008. 2. Factories with 10 or more workers using power are covered. 3. GVA per person engaged grew faster than corporate GVA in 2024–25. Which are correct? **(Ans: 1 and 2 only)**

FQ – MAINS (Possible Question)

- "India's factory sector shows growth without broad-based productivity." Analyse using ASI 2024–25. (250 words)

GS-III | Economy – Infrastructure (Energy); Renewable Energy UPDATE

9. ₹14,500-Crore Push for 92 GWh of Grid Energy Storage

Syllabus: *GS-III: Infrastructure – energy; conservation; government schemes for renewable energy.*

WHAT WE KNOW TILL NOW

- India's **solar boom** means power is plentiful **by day** but scarce **at night** – **storage** is the missing link.
- **PM Surya Ghar: Muft Bijli Yojana** funds **rooftop solar for 1 crore homes**; the CEA sees a big **battery need by 2031–32**.
- Earlier **viability gap funding** supported **battery storage** projects.

WHAT TODAY ADDS

- A new **₹14,500-crore VGF scheme** for **92 GWh** of storage – **pumped hydro, batteries** and **new technologies**.
- **Railways** topped the **PM Surya Ghar Excellence Award 2026** for **rooftop solar on government buildings**.

CONTEXT

A **Viability Gap Funding (VGF)** scheme worth **₹14,500 crore** (**2026–27 to 2030–31**) got **Cabinet** approval for **92 GWh of grid-scale energy storage**; separately the **Ministry of Railways** won the **PM Surya Ghar Excellence Award 2026** for **solarising the most buildings**.



Illustration: a pumped-storage plant with two reservoirs, battery containers beside solar panels and wind turbines, and a solar-roofed railway station.

KEYWORDS DEFINED

Keyword	Meaning
VGF	**Viability Gap Funding** – a **one-time capital grant** that makes a needed but **not-yet-profitable** project bankable.
GWh	**Gigawatt-hour** – a unit of **stored energy** (how much, not how fast).
PSP	**Pumped Storage Plant** – pumps water **uphill** when power is cheap and

Keyword	Meaning
	releases it through turbines at peak.
BESS	**Battery Energy Storage System** – large batteries that **store and release** grid power.
DCR	**Domestic Content Requirement** – a minimum share of **locally made parts**.

The storage scheme

- **₹14,500 crore** VGF over **five years (FY27–FY31)**; expected to pull in about **₹97,600 crore** of investment.
- **Mix: 60 GWh pumped storage**, **30 GWh batteries (BESS)**, **2 GWh** for **new technologies**.
- Adds about **170%** of today's grid storage, towards a **411 GWh need by 2031–32**.
- Awarded by **tariff-based competitive bidding** – lowest storage tariff, least support wins.

Design features

- **40% domestic content** for VGF-backed **battery projects** – boosts **Aatmanirbhar** battery making.
- **Public and private DISCOMs** can get VGF for **BESS and pumped storage**.
- Helps deliver **round-the-clock (RTC) green power**, steadies the **grid** and cuts **curtailment** of solar and wind.

Railways' rooftop record

- Solar panels on **12,495 of 15,868** identified railway buildings – **79% coverage**, **first among Central ministries**.
- **351 MW** commissioned against a **424 MW** target (about **83%**).
- Railways make up about **48%** of the **25,984 government buildings** solarised across **77 ministries and departments**.
- Aim: **all feasible railway buildings** solarised by **March 2027**; the award is given by **MNRE**.

Why it matters

- Stores **midday solar** for the **evening peak**, reducing reliance on **coal**.
- Builds a **battery supply chain** alongside **PLI for Advanced Chemistry Cells**.
- **Value addition:** Indian Railways aims to be **net-zero carbon by 2030**; India targets **500 GW non-fossil capacity by 2030** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **storage technologies**, **VGF**, **PM Surya Ghar** and **renewable targets**.

PYQ – MAINS

- **Pattern:** Mains asks how to **integrate renewables** into the grid and manage **intermittency**.

FQ – PRELIMS (Possible Question)

- Under the new VGF scheme for 92 GWh of grid storage, the largest share is for: (a) battery storage (b) pumped storage (c) green hydrogen (d) compressed air storage **(Ans: b)**

FQ – MAINS (Possible Question)

- "Without storage, more solar means more wasted power." Discuss India's storage strategy. (150 words)

GS-III | Science & Technology – Nuclear Fusion **UPDATE**

10. How a Tokamak Tames the Sun's Fire

Syllabus: *GS-III: Science and technology – developments and applications; energy; indigenisation of technology.*

WHAT WE KNOW TILL NOW

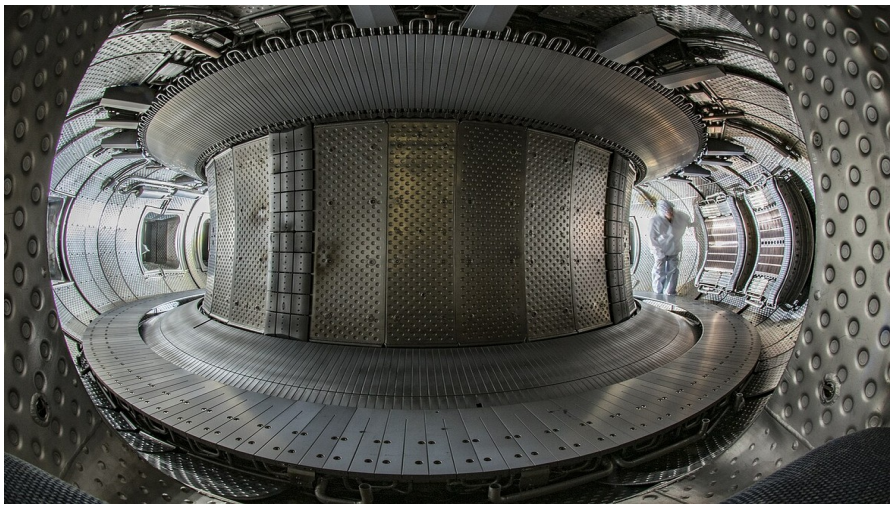
- A Bengaluru start-up, **Pranos Fusion**, built **PRAGYA**, a **compact research tokamak**, and plans a **commercial reactor design**.
- India's public tokamaks **ADITYA** and **SST-1** are at the **Institute for Plasma Research, Gandhinagar**; India is an **ITER** member.

WHAT TODAY ADDS

- **How a tokamak works** – plasma, twisted magnetic fields and the **D–T reaction**.
- Why the **doughnut shape**, the hard problems, and **rival designs** like stellarators and lasers.

CONTEXT

With global interest in **fusion as clean energy** rising, the **tokamak** – a **doughnut-shaped magnetic bottle** for super-hot plasma – is the leading design for **commercial fusion power**.



Inside the WEST tokamak in France: the ring-shaped steel vacuum chamber where plasma is held by magnetic fields.
Photo: Christophe Roux / IRFM, Wikimedia Commons, CC BY 4.0

KEYWORDS DEFINED

Keyword	Meaning
Tokamak	A **ring-shaped chamber** where **strong magnetic fields** hold **plasma** for **controlled fusion**.
Plasma	The **fourth state of matter** – gas so hot that **electrons are stripped** from nuclei.
Deuterium–Tritium (D–T)	Two **hydrogen isotopes** that fuse into **helium and a fast neutron**.
Blanket	A layer around the plasma that **absorbs neutron heat** and can **breed tritium from lithium**.

Keyword	Meaning
Stellarator	A fusion device whose **twisted coils** make the field **without a plasma current**.

Fusion basics

- **Fusion:** two **light nuclei join** into a heavier one and release **huge energy** – the Sun's process.
- **D–T fusion** gives **helium plus a neutron** and about **17.6 MeV** – **80% with the neutron**, **20% with helium**.
- Fully fused, a kilogram of D–T fuel holds about **ten million times** the energy of the same mass of **coal**.

How the tokamak works

- Gas is heated until it becomes **plasma** – **hotter than the Sun's core**.
- **Magnetic fields** keep plasma **off the walls**; a **ring** removes the **open ends** of a straight tube.
- A plain ring field is **stronger inside than outside**, so particles **drift** and plasma is pushed **outwards**.
- A **current driven through the plasma** adds a second field; together they make a **twisted (helical) field** that holds plasma stable.
- **Neutrons** heat the **blanket** → steam → turbines; **lithium** in the blanket can **breed tritium**.

Hurdles and alternatives

- **Instability**, **extreme heat**, **neutron damage to materials**, **scarce radioactive tritium**, **high cost**.
- No tokamak has yet produced **net electricity**.
- **Stellarators** (e.g., **Wendelstein 7-X**, Germany) – steadier but complex.
- **Inertial confinement:** lasers crush a fuel pellet – the US **National Ignition Facility** reached **ignition in 2022**.

Global and Indian progress

- **ITER**, **Cadarache, France** – members **China, EU, India, Japan, Korea, Russia, US**; aims for **500 MW from 50 MW input**; **not for electricity**.
- **China's EAST** (Hefei) held high-confinement plasma for **1,066 seconds** in **January 2025**.
- **India: ADITYA (1989) → ADITYA-U (2017)**; **SST-1** first plasma **2013** with **superconducting magnets**; private **PRAGYA** by **Pranos Fusion**.
- **Value addition:** India joined **ITER in 2005**; **ITER-India** works under the **IPR** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **ITER**, **fusion vs fission**, and **Indian research facilities**.

PYQ – MAINS

- **Pattern:** Mains asks about **clean energy technologies** and India's **science capacity**.

FQ – PRELIMS (Possible Question)

- Consider: 1. ITER is designed to supply electricity to the French grid. 2. A tokamak uses a plasma current to help create a twisted magnetic field. 3. Lithium in the blanket can breed tritium. Which are correct? **(Ans: 2 and 3 only)**

FQ – MAINS (Possible Question)

- "Fusion is always thirty years away." Examine the technical hurdles and India's role in fusion research. (150 words)

GS-III | Internal Security – Defence Manpower; Governance UPDATE

11. Revamped Agnipath: More Chances, Not a Return to Old Recruitment

Syllabus: *GS-III: Security challenges and their management; various security forces and their mandate. GS-II: Government policies.*

WHAT WE KNOW TILL NOW

- **Agnipath (2022)** recruits **Agniveers** for **four years**; originally only **25%** were kept as regulars.
- The **Cabinet** has approved a **two-year second engagement** for the other **75%**, with higher pay and a larger **Seva Nidhi**.

WHAT TODAY ADDS

- **Absorption is a chance, not a guarantee** – selection after six years still needs **willingness and objective criteria**.
- Those taking the extension **lose reservation benefits** in **CAPFs, PSUs and State jobs**; **seniority** and **scale** details.

CONTEXT

After the **4 November 2026** Cabinet decision, the **Chief of Defence Staff, General N.S. Raja Subramani**, said **Agnipath continues** with changes – a **second chance at regular service**, but **not a return** to the **pre-COVID system**.



Illustration: a young soldier on a road that forks towards a military camp and a city, with a calendar showing extra years.

KEYWORDS DEFINED

Keyword	Meaning
Agniveer	A soldier recruited under **Agnipath** for a **fixed term**, not as a permanent soldier from day one.
Seva Nidhi	An **exit corpus**: **30% of pay** plus an **equal government share**, about **₹11.71 lakh** after four years, **tax-free**.

Keyword	Meaning
Regular cadre	**Permanent soldiers** with full **service benefits**, including **pension** under rules.
Two-stage selection	**25% chosen at four years**; the rest get **another chance at six years**.
CDS	**Chief of Defence Staff** – head of the **Department of Military Affairs**.

What changes

- **25%** still chosen **on merit** for permanent service **after four years**.
- The other **75%** may take a **two-year extension**, then get **another chance** to join as **regulars** – if **willing and selected**.
- **Enhanced pay** in the extension and an **enhanced Seva Nidhi** for those who leave at **six years**.
- **Trade-off:** those who take the extension **lose quota benefits** in **paramilitary forces, PSUs and State jobs**.

Why it is still not old-style recruitment

- Old system: **regular soldier from day one**; Agnipath: **fixed term first**, permanence **earned later**.
- Permanence comes via **two selection stages**, not at entry.
- Leavers at four or six years get **Seva Nidhi** but **no pension or gratuity**.
- **Seniority:** six-year absorbees rank **just below** their batchmates absorbed at four years.

Benefits for the forces

- Eases **manpower gaps** from the **COVID-era recruitment pause** and normal retirements.
- Keeps **trained, experienced** soldiers while holding a **young profile**.
- Competition at each stage may **reward merit**; trained leavers add **skills to the civilian economy**.
- Important after **Galwan** and **Operation Sindoor** for **operational readiness**.

Scale

- **Over 3 lakh** Agniveers in training or service across three services; about **2.8 lakh in the Army**, of whom about **19,000** finish four years first.
- About **2,500 Navy** Agniveers finish in **November 2026**; **2,900** in the **first IAF batch** in **December 2026**.
- **Value addition:** the CDS post was created in **2019**; **General Bipin Rawat** was the first CDS *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **defence reforms**, **CDS** and **recruitment schemes**.

PYQ – MAINS

- **Pattern:** Mains asks about **military modernisation, manpower** and **civil–military** questions.

FQ – PRELIMS (Possible Question)

- Under the revamped Agnipath scheme: 1. 25% of Agniveers are still selected after four years. 2. All Agniveers are guaranteed permanent absorption after six years. 3. Agniveers taking the extension lose reservation benefits in CAPFs. Which are correct? **(Ans: 1 and 3 only)**

FQ – MAINS (Possible Question)

- "The Agnipath revamp changes retention, not the recruitment philosophy." Examine. (150 words)

GS-III | Environment – Climate Change Mitigation; Transport

12. NDC-TIA II: India and Germany Team Up for Electric Trucks

Syllabus: *GS-III: Environment – climate change mitigation; infrastructure – roads and logistics. GS-II: Bilateral agreements.*

CONTEXT

NITI Aayog and Germany's **GIZ** signed an **implementation agreement** under **Phase II of the NDC Transport Initiative for Asia (NDC-TIA II)** to speed up **electric freight** in India.

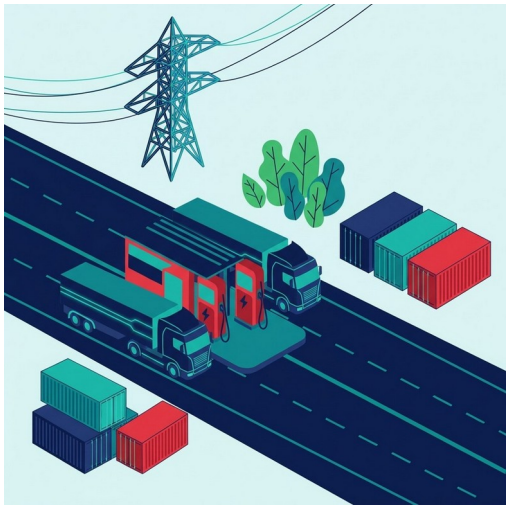


Illustration: electric trucks on a freight highway at a high-power charging station, with a power tower, leaves and cargo containers.

KEYWORDS DEFINED

Keyword	Meaning
NDC	**Nationally Determined Contribution** – a country's **climate pledge** under the **Paris Agreement**.
GIZ	**Deutsche Gesellschaft für Internationale Zusammenarbeit** – Germany's **development cooperation agency**.
IKI	Germany's **International Climate Initiative**, which **funds** NDC-TIA.
TCO	**Total Cost of Ownership** – the **full lifetime cost** of a vehicle, not just its price.
MHDV	**Medium- and heavy-duty vehicles** – trucks that emit far more than their **small share of the fleet**.

About the initiative

- A **regional cooperation programme** to cut **transport emissions** and help countries meet their **NDCs**.
- **Countries: India, China and Vietnam**; funded by **Germany's IKI** (about **€4.04 million** for the regional programme).
- **Phase I** began around **2019–20**; **Phase II** runs from **November 2024 to September 2028**.
- **Coordinated by GIZ**; in India **NITI Aayog** leads, with **WRI India, ICCT India** and **Agora Energiewende**.

What it will do

- Push **electric medium and heavy trucks** – the big polluters in freight.

- Plan **high-power charging** on **freight corridors** and the **grid upgrades** they need.
- Use **blended finance, risk-sharing** and **TCO-based** models to make **zero-emission trucks** affordable.
- Bring **ministries, truck makers, logistics firms and power utilities** together; share **data and lessons** across the three countries.

Why freight matters

- Trucks move most of India's goods by road; **diesel trucks** are a big source of **CO_2 and PM2.5**.
- Electric trucks can cut **fuel imports** and **city pollution**.
- **Hurdles: battery cost, range, charging time** and **grid capacity**.

Indian links

- **PM E-DRIVE** supports **e-trucks and charging**; **Gati Shakti** plans **multimodal logistics**.
- **Value addition:** India's updated NDC targets **45% lower emissions intensity of GDP by 2030 (vs 2005)** and **net zero by 2070** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **climate initiatives, their partners and funders** and **Paris Agreement** terms.

PYQ – MAINS

- **Pattern:** Mains asks about **decarbonising transport** and **green finance**.

FQ – PRELIMS (Possible Question)

- NDC-TIA II, recently in the news, covers which countries? (a) India, Bangladesh, Nepal (b) India, China, Vietnam (c) India, Indonesia, Thailand (d) India, Japan, Korea **(Ans: b)**

FQ – MAINS (Possible Question)

- "Decarbonising road freight is the hardest part of India's transport transition." Discuss. (150 words)

GS-III | Internal Security – Defence Indigenisation (Navy)

13. INS Amini: A Home-Built Submarine Hunter for Shallow Seas

Syllabus: *GS-III: Security challenges; indigenisation of technology; defence production.*

CONTEXT

The **Indian Navy** received **'Amini'**, an **Anti-Submarine Warfare Shallow Water Craft (ASW SWC)** designed and built in India by **GRSE, Kolkata**, on **5 November 2026**.



Illustration: a small grey warship on waterjets near a palm-fringed island, with sonar waves reaching a submarine outline below.

KEYWORDS DEFINED

Keyword	Meaning
ASW	**Anti-submarine warfare** – finding and attacking **enemy submarines**.
SWC	**Shallow Water Craft** – a small warship made for **coastal, littoral waters**.
Waterjet propulsion	Moves a ship by **shooting water jets** instead of propellers – good in **shallow water**.
Sonar	Uses **sound waves** to **detect objects under water**.
GRSE	**Garden Reach Shipbuilders & Engineers, Kolkata** – a **defence PSU** shipyard.

About Amini

- An **ASW Shallow Water Craft**, **designed and built in India** by **GRSE, Kolkata**.
- Named after **Amini Island** in **Lakshadweep**; revives the name of the earlier **INS Amini**, a **Petya III-class** corvette **retired in 2002**.
- Over **80% indigenous content**.

Capabilities

- Among the **largest Indian warships** driven by **waterjets**.
- Carries **lightweight torpedoes**, **indigenous anti-submarine rockets** and **sonar**.
- Roles: **anti-submarine warfare**, **coastal surveillance** and **mine-laying**.

Why it matters

- Guards **shallow coastal waters**, ports and approaches where large ships work poorly.
- Counters growing **submarine activity** in the **Indian Ocean Region**.
- Shows **Aatmanirbhar** shipbuilding – part of a **16-ship ASW SWC programme** shared by **GRSE and Cochin Shipyard** *(general knowledge)*.

Map and context

- **Lakshadweep:** a coral archipelago in the **Arabian Sea**; **Kavaratti** is the capital; **Amini** is one of its inhabited islands.
- **Value addition:** the **Arnala** was the **first ship** of this class, commissioned in **2025** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **naval vessels, shipyards** and **islands** they are named after.

PYQ – MAINS

- **Pattern:** Mains asks about **maritime security** and **defence indigenisation**.

FQ – PRELIMS (Possible Question)

- The ASW Shallow Water Craft 'Amini' is named after an island in: (a) Andaman and Nicobar Islands (b) Lakshadweep (c) the Gulf of Mannar (d) the Sundarbans **(Ans: b)**

FQ – MAINS (Possible Question)

- "Littoral security needs small, smart ships." Discuss with reference to India's shallow-water ASW craft. (150 words)

GS-III | Science & Technology – Space Debris and the Private Space Sector

14. Cosmoserve's Base 1: India's Private Bet on Clearing Space Junk

Syllabus: *GS-III: Awareness in the field of space; indigenisation of technology.*

> **CONTEXT**
>
> A **Union Minister** opened **Cosmoserve Space's Base 1** in **Hyderabad**, a facility for **Active Debris Removal (ADR)** and **In-Orbit Servicing (IOS)**, supporting India's goal of **Debris-Free Space by 2030**.



Illustration: a small servicing spacecraft using soft gripper petals to catch a tumbling dead satellite above the Earth.

KEYWORDS DEFINED

Keyword	Meaning
Space debris	**Dead satellites, spent rocket stages and fragments** left in orbit.
Kessler Syndrome	A **chain reaction of collisions** that could make some orbits **unusable**.
ADR	**Active Debris Removal** – catching and **de-orbiting** junk.
IOS	**In-Orbit Servicing** – **refuelling, repairing or inspecting** satellites in space.
IN-SPACe	**Indian National Space Promotion and Authorisation Centre** – authorises and promotes **private space activity**.

The facility

- A **14,800 sq ft** centre in **Hyderabad** with **robotics labs**, **microgravity simulation**, **clean rooms** and a **mission control** room.
- Built by **Cosmoserve Space**, an **Indian private start-up**, under the **IN-SPACe, DoS and ISRO** framework.

Technology

- **Soft robotic gripper 'petals'** catch **tumbling, odd-shaped** objects **without making new fragments** – tested in **Mission EMBRACE (July 2026)**.
- Partners: **Skyroot Aerospace** (launch), **Pixxel** (satellite systems), **Dhruva Aerospace Netherlands** (propulsion).
- Offers both **debris removal** and **commercial servicing** – **refuelling, inspection, assembly, life extension**.

India's debris effort

- **ISRO's DFSM** (Debris-Free Space Missions) – **no new debris** from Indian missions by **2030**.
- **Project NETRA** – ISRO's **space-object tracking** network.
- **Shift** from **tracking debris** to **removing it**.

Why it matters

- Protects **satellites** that India relies on for **communication, navigation and weather**.
- Opens a **new market** as India aims to grow its **space economy to $40–45 billion**.
- **Value addition:** the **Indian Space Policy 2023** opened all space activities to **private players** (*general knowledge*).

PYQ – PRELIMS

- **Pattern:** Prelims asks about **space agencies, programmes** and terms like **Kessler Syndrome**.

PYQ – MAINS

- **Pattern:** Mains asks about the **private space sector** and **space sustainability**.

FQ – PRELIMS (Possible Question)

- Project NETRA of ISRO is related to: (a) deep-space communication (b) space situational awareness and debris tracking (c) a navigation satellite system (d) a crewed mission **(Ans: b)**

FQ – MAINS (Possible Question)

- "Space sustainability is the next frontier of space policy." Discuss with reference to India's debris-free goal. (150 words)

GS-III | Science & Technology – Research and Innovation Policy

15. India S&T Scoreboard: Mapping Where Research Happens

Syllabus: *GS-III: Science and technology – developments; indigenisation. GS-II: Federalism; government policies.*

CONTEXT

An **interactive dashboard** of research, patents and innovation for **all States and UTs** – the **India Science and Technology (S&T) Scoreboard** – has been rolled out by **NITI Aayog**.



Illustration: a large analytics dashboard with charts and a ranking table beside icons for papers, patents, a microscope, an atom and a bulb.

KEYWORDS DEFINED

Keyword	Meaning
S&T Scoreboard	**NITI Aayog's dashboard** that brings together **research, patent and innovation data** by State.
CNCI	**Category Normalised Citation Impact** – citations compared with the **world average of 1.0**.
Q1 journal	A journal in the **top quarter** of its field by citations.
Competitive federalism	States **compete** to improve on **public indicators**.
Patent	A legal right to an **invention** for a fixed time, in return for **disclosing it**.

About the Scoreboard

- Built by **NITI Aayog's S&T Division** with data from sources such as **Web of Science (Clarivate)**.
- Covers **all 36 States and UTs**, **FY 2021–22 to FY 2024–25**, with **universities, labs, start-ups and incubators**.
- Splits papers and patents by **academia, government R&D labs and industry**.
- Tracks **11 priority areas** – including **AI, semiconductors, clean energy, critical minerals, biotech, space**.

Quality indicators

- **Top 10% most-cited papers: 8.5%** of India's output.
- **Q1 journal papers: 28.5%**; **international collaboration: 29.9%**.
- **CNCI: 0.89** – below the **world average of 1.0**.

What it shows

- Research is concentrated in **Tamil Nadu, Uttar Pradesh, Maharashtra, Karnataka and Delhi**.
- Lets States **compare** research, patents and policies and copy **best practices**.
- Aligns State science with **Viksit Bharat 2047**.

Why it matters

- Helps target **funding gaps** and **institution building**.
- Links **labs with industry** for **technology transfer** and start-ups.
- **Value addition:** India's **gross R&D spending is about 0.6–0.7% of GDP**; the **Anusandhan National Research Foundation Act, 2023** set up **ANRF** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **NITI Aayog indices** and **research bodies** like ANRF.

PYQ – MAINS

- **Pattern:** Mains asks why India's **R&D spend is low** and how to build **innovation ecosystems**.

FQ – PRELIMS (Possible Question)

- In the India S&T Scoreboard, India's Category Normalised Citation Impact is about 0.89. This means: (a) above the world average (b) below the world average of 1.0 (c) equal to the world average (d) 89% of papers are cited **(Ans: b)**

FQ – MAINS (Possible Question)

- "States, not just the Centre, must drive India's research ambitions." Discuss. (150 words)

GS-III | Environment – Protected Areas and Species Reintroduction

16. Blackbucks Return to Gomarda After Five Decades

Syllabus: *GS-III: Conservation, environmental degradation; protected areas. GS-I: Geography of India (places in news).*

CONTEXT

Chhattisgarh will bring **blackbucks back to Gomarda Wildlife Sanctuary**, where they vanished in the **1970s** – **50 animals** will come from **Barnawapara Wildlife Sanctuary**.



A blackbuck stag with long spiral horns in dry grassland in Gujarat.
Photo: Chinmayisk, Wikimedia Commons, CC BY-SA 3.0

KEYWORDS DEFINED

Keyword	Meaning
Blackbuck	*Antilope cervicapra* – an **antelope of open grasslands**; males have **spiralled horns** and dark coats.
Reintroduction	Bringing a species **back to an area where it died out**.
Translocation	Moving animals **from one wild site to another**.
Tropical dry deciduous forest	Forests that **shed leaves in the dry season**; rain about **700–1,500 mm**.
Schedule I	**Highest protection** under the **Wildlife (Protection) Act, 1972**.

About Gomarda

- In **Raigarh district, Chhattisgarh**; notified in **1975**; about **278 sq km**.
- Ringed by hills – **Rathan, Budhaghat, Gomarda, Danav Karwat**; rocky, rolling land unfit for farming, so **naturally protected**.
- Rivers **Laat and Manai**; waterfalls **Adhar Paani, Khapan, Madosilli**; rain about **1,400–1,500 mm**.

Forests and wildlife

- **Tropical dry deciduous** and mixed forest – **sal, teak, bamboo, tendu, mahua, bija, saja**.
- Animals: **sambar, gaur, barking deer, chinkara, sloth bear, wild boar, dhole, jackal**.

The reintroduction

- **50 blackbucks** to be moved from **Barnawapara WLS**.

- They vanished from the area by the **1970s** due to **hunting, poaching, encroachment and development**.
- Success needs **open grassland**, **predator and poaching control** and **community support**.

About the blackbuck

- **IUCN: Least Concern**; **Schedule I** of the **Wildlife (Protection) Act**.
- **State animal** of **Punjab, Haryana and Andhra Pradesh**.
- Protected by the **Bishnoi community** of Rajasthan; **Velavadar Blackbuck National Park** is in **Gujarat** *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **sanctuaries and their States**, **species status** and **State animals**.

PYQ – MAINS

- **Pattern:** Mains asks about **species recovery** and **grassland conservation**.

FQ – PRELIMS (Possible Question)

- Gomarda Wildlife Sanctuary, recently in the news for blackbuck reintroduction, is located in: (a) Madhya Pradesh (b) Odisha (c) Chhattisgarh (d) Jharkhand **(Ans: c)**

FQ – MAINS (Possible Question)

- "Reintroductions succeed only when the causes of local extinction are fixed." Discuss with an example. (150 words)

GS-III | Environment – Biodiversity (New Species)

17. Disporum kamakhasiense: A New Flowering Plant of the North East

Syllabus: *GS-III: Conservation; biodiversity. GS-I: Geography – North-East India.*

CONTEXT

Disporum kamakhasiense, a **new species of flowering plant**, has been found in forests of **Assam and Meghalaya**.



Illustration: a tall herb with branched stems, broad veined leaves and small white star-shaped flowers on a cliff under bamboo.

KEYWORDS DEFINED

Keyword	Meaning
Perennial herb	A **non-woody** plant that **lives for more than two years**.
Colchicaceae	A **flowering-plant family** that includes *Disporum* and *Gloriosa* (the flame lily).
Endemic	Found **naturally in one region only**.
Type locality	The place where a **new species is first collected** and described.
Indo-Burma hotspot	One of India's **four biodiversity hotspots**, covering most of the **North East**.

About the plant

- A **perennial herb** of the **Colchicaceae** family.
- Found at **three sites**: **Garbhanga Reserve Forest (Assam)**, **Laisong in Dima Hasao (Assam)** and **Barapani (Meghalaya)**.
- Name joins **Kamarupa** (old name of Assam) and the **Khasi Hills** (Meghalaya).

Features and habitat

- Grows on **cliffs**, mostly **under bamboo forest**, at **200–1,100 m**.
- **Branched stems**, **broad yellowish-green veined leaves**, **white star-shaped flowers**; up to **180 cm** tall.
- Told apart by **longer flower stalks** and **shorter, thicker stamens**.

Conservation

- Described as **'potentially endangered'**.
- Threats: **road widening, over-harvesting and deforestation**.

Why it matters

- Shows how much of the **North East's flora** is still **undescribed**.
- **Value addition:** India's four hotspots are the **Himalaya, Indo-Burma, Western Ghats–Sri Lanka and Sundaland (Nicobar)**; the **Botanical Survey of India (1890)** documents plant life *(general knowledge)*.

PYQ – PRELIMS

- **Pattern:** Prelims asks about **new species**, their **States** and **biodiversity hotspots**.

PYQ – MAINS

- **Pattern:** Mains asks about **biodiversity loss from infrastructure** in fragile regions.

FQ – PRELIMS (Possible Question)

- Disporum kamakhasiense, recently discovered, was recorded in: (a) Arunachal Pradesh and Nagaland (b) Assam and Meghalaya (c) Sikkim and West Bengal (d) Mizoram and Tripura **(Ans: b)**

FQ – MAINS (Possible Question)

- "New species are being found faster than we can protect them." Discuss with reference to North-East India. (150 words)

GENERAL STUDIES PAPER IV

[GS-IV] Ethics, Integrity & Aptitude

GS-IV | Ethics, Integrity & Aptitude – Synthesis

18. Ethics Synthesis: Protecting Without Excluding, Pricing With Conscience

Syllabus: *GS-IV: Ethics in public administration; probity; ethical issues in technology and health; case studies.*

CONTEXT

Today's topics raise three public-service values: **protection without exclusion** (children online), **fairness and compassion in health** (cancer drug prices), and **integrity of public institutions** (State PSCs).



Illustration: a balance with a child holding a tablet on one side and a medicine bottle on the other, a shield in the middle and a certificate below.

KEYWORDS DEFINED

Keyword	Meaning
Proportionality	A restriction must be **no more than needed** to meet a **legitimate aim**.
Paternalism	Limiting someone's choices **'for their own good'**, without their consent.
Distributive justice	Fair sharing of **scarce goods** such as **medicines**.
Conflict of interest	When **personal gain** could sway an **official duty**.
Institutional integrity	An institution acting **fairly, openly and by its own rules**.

The ethical thread

- **Protection vs freedom:** the Court chose **safeguards over bans** – **proportionality**, not **paternalism**.
- **Compassion and justice:** a lower **MRP** means little if **hospitals keep the margin** – patients need **real savings**.
- **Integrity:** a **PSC** is trusted only if its members are **above suspicion** and removal follows **due process**.
- **Evidence: ASI** data show where **workers lag**; good policy follows **data, not slogans**.

Values at stake

- **Proportionality, compassion, fairness, transparency, integrity, accountability, empathy, objectivity.**

Case study 1: The school phone ban

- **Situation:** As **District Collector**, parents demand you **ban phones and internet** for all under-16s in schools after a **cyberbullying** case.
- **Issues: child safety vs right to learn**; **public pressure vs proportionality**.
- **Options:** (a) impose a blanket ban; (b) do nothing; (c) set **age-appropriate rules**, **filtered school networks**, **counsellors and a reporting cell**, and **online-safety classes** with parents.
- **Preferred course: (c)** – **protect without excluding**; children who fear punishment **hide abuse**.

Case study 2: The hospital pharmacy

- **Situation:** As **medical superintendent** of a trust hospital, you find the pharmacy **stocks costly brands** because a **flat 30% margin** earns more, while **quality generics** exist.
- **Issues: revenue vs patient welfare**; **conflict of interest**.
- **Options:** (a) continue for income; (b) stop all branded drugs; (c) **stock quality generics first**, **show prices openly**, give **itemised bills**, and let doctors **explain choices**.
- **Preferred course: (c)** – **beneficence and transparency** without lowering **quality**.

Case study 3: The PSC member

- **Situation:** As a **PSC member**, you learn a **relative** is a candidate in an interview board you sit on.
- **Issues: conflict of interest**; **public trust** in recruitment.
- **Options:** (a) sit and be fair; (b) stay silent; (c) **declare the interest in writing** and **recuse** from that board.
- **Preferred course: (c)** – **justice must be seen to be done**.

Thinkers to quote (paraphrased)

- **John Stuart Mill:** power over a person is justified **only to prevent harm to others**.
- **Amartya Sen:** development is the **expansion of capabilities** – access to knowledge is one.
- **Mahatma Gandhi:** recall the **poorest face** before acting.
- **Kautilya:** in the happiness of the subjects lies the **king's happiness**.
- **Lord Hewart:** justice must not only be done but **seen to be done**.

PYQ – PRELIMS

- **Not applicable** – GS-IV is Mains-only.

PYQ – MAINS

- **Pattern:** GS-IV asks about **conflict of interest**, **ethics in technology** and **compassion in public service**.

FQ – PRELIMS (Possible Question)

- **Not applicable** – GS-IV is Mains-only.

FQ – MAINS (Possible Question)

- "Protection that excludes is not protection." Discuss with reference to children's digital rights. (150 words)
- Case study: As a medical superintendent finding costly brands preferred over generics, what would you do and why? (250 words)